COMP 018: Database Administration - Activity 6 Progress Report
Database Backup Planning, Verification, Restoration, and Disaster Recovery Simulation

Project Identification & Metadata

Course	: COMP 018 - Database Administration
Evaluating Group	: Group 4: Olivarez, Charles Kean E., Infantado, Gerwin G., Pocsidio, Jhon Louie F., Abangan, SJ Kaizen M., Bartolata, Redge Raindel N.
Evaluating Group Repository	: https://github.com/Nycri/brewpos
Target Project Monitored	: HabalGo: Ride and Booking Management System
Target Project Group	: Group 6 (Royo, Villareal, Barte, Balansay, Remoquillo, Gumahin)
Target Project Repository	: https://github.com/jeanaroyo/HabalGo
Primary Database	: grp00_habalgodb (PostgreSQL 18.6 on x86_64 Windows, compiled by MSVC)
Target Schema	: habalgo
Restored Test Database	: grp00_habalgodb_restore_test

ACTIVITY 6: Database Backup, Restoration, and Recovery

3.1 Backup Planning & Architecture

The disaster recovery plan for the HabalGo database encompasses eight relational tables within the habalgo schema, capturing customer profiles, driver fleet data, vehicle assignments, location entries, bookings, payment records, and user ratings. The plan covers sequence counters, primary keys, foreign keys, unique constraints, and privilege grants to ensure operational continuity.

The backup strategy utilizes PostgreSQL custom compressed archive format via a dedicated backup operator role. Archive dumps are stored in a protected directory outside web roots and repository trees. Recovery objectives define a Recovery Point Objective of 24 hours and a Recovery Time Objective of 2 hours. Retention guidelines specify maintaining seven daily backups alongside four weekly snapshots.

Because database login accounts exist at the PostgreSQL cluster level and are not stored within single-database dumps, disaster recovery plans require executing access-control role scripts on new server instances prior to restoring database contents.

3.2 Backup Creation, Checksum & Archive Inspection

Logical backup creation was executed using the dedicated backup account, producing a compressed custom format archive sized at 32,794 bytes. Cryptographic verification computed a SHA-256 hash value of 6EA2832EB93D77B87A93DA7AEEC44BC377E8DD14633D62632239EA292CC6F549,

establishing a baseline for integrity verification.

Inspection of the archive table of contents verified 94 catalog entries, confirming the inclusion of all eight database tables, sequence objects, primary keys, check constraints, indexes, and object privilege grants. To ensure recoverability beyond file inspection checks, a complete restoration drill was conducted in an isolated test database.

3.3 Restoration into Separate Test Database
An isolated test database named grp00_habalgodb_restore_test was created on the local server instance. The custom archive dump file was restored into this database using administrative utility tools. The restoration process completed cleanly with zero error codes.

3.4 Post-Restore Verification Suite
Post-restoration verification confirmed that all eight relational tables were recreated with proper schema structure under administrative ownership. Record count validation across tables demonstrated complete parity with the source database, matching two customers, ten drivers, four vehicles, three driver-vehicle assignments, six locations, six bookings, four payment records, and two rating feedback entries. Database checks confirmed that all 58 constraints and 17 indexes were fully restored.

Access control verification on the restored test database confirmed that privilege definitions transferred accurately. Queries executed under the runtime user account successfully returned booking records, while insertion attempts by the read-only auditor account and deletion attempts on payment records were blocked by the database security layer.

3.5 Controlled Recovery Simulation
A controlled data recovery simulation was performed to test incident response procedures. A disposable booking record was inserted into the restored test database to establish a baseline state. A data loss event was then simulated by removing the record, and queries confirmed its deletion.

Targeted record restoration was executed to reinsert the missing data row without affecting existing database entries. Post-recovery verification confirmed that the record was fully restored with its original attributes and timestamps intact. Following verification, the test record was removed to clean up the test environment.

Scenario evaluation noted that standard logical backups recover data up to the snapshot timestamp, meaning transactions created between the backup and an incident fall within the 24-hour RPO window. For production systems requiring near-zero data loss, continuous write-ahead log archiving and point-in-time recovery are required.

3.6 Limitations, Evaluation & Production Recommendations
Restoration procedures on fresh server installations require pre-executing role definition scripts

to establish user accounts before restoring database ACLs. For production operations, daily automated backup generation should be scheduled using system task schedulers to archive database dumps with date-stamped filenames and retain snapshots according to policy. For environments requiring near-zero data loss tolerances, enabling continuous write-ahead log archiving will support point-in-time recovery capabilities.

CLASS GOOGLE SHEETS TRACKER ENTRY (Activity 6)

Column Header	**Value**
Project	Group 6 - HabalGo: Ride and Booking Management System
Database and backup file	grp00_habalgodb, grp00_habalgodb_activity6.dump, Size: 32,794 bytes, SHA256:
Backup method	pg_dump custom format (-Fc), compressed binary dump (2026-11-05 06:46:58)
Restore test result	Success (Restored into grp00_habalgodb_restore_test; all 8 tables, sequences, constraints, and data verified)
Access control after restore	(Activity 4 roles and permissions validated; habalgo_readonly_user denied inserting)
Recovery simulation result	Recovered (Simulated deleted test booking successfully restored from baseline state)
Proposed RPO and RTO	RPO: 24 hours, RTO: 2 hours
Evidence folder link	https://github.com/Nycri/brewpos/tree/main/activity-6/Evidence-act6
Feedback sent to owners	(05/11/2026)
PDF progress document	Activity6_Progress_Report_Group6.pdf
Last updated	2026-11-05

AI USAGE DECLARATION

Google Gemini 3.8 Flash was utilized during project evaluation to inspect database schema

definitions, evaluate relational hierarchy structures, formulate role verification test scenarios, analyze database maintenance operations and query performance execution plans, verify backup creation and restoration procedures, and structure technical report documentation. All database statements, configuration checks, and verification procedures were directly executed and validated against PostgreSQL 18.6 within the local database environment.